

# Datasheet — VBF-T6R2-DSH-001

## Datasheet — LNMO/Graphite 4C Smartphone Cell (VBF-T6R2-DSH-001)

**\*\*Case\*\*:** exp/t6\_r2 · **\*\*Candidate\*\*:** R12a\_h140 · **\*\*Date\*\*:** 2026-08-26  
All values are tool-output sourced (DFN precise model unless noted).

### Electrical

| Item                                   | Value                   | Source                         |
|----------------------------------------|-------------------------|--------------------------------|
| Nominal capacity                       | 6.374 Ah (1C discharge) | cell/r12_a_1c_dfn.json:capacit |
| Discharge energy                       | 26.52 Wh                | cell/r12_a_energy_dfn.json:en  |
| Volumetric energy density              | 1096.0 Wh/L             | cell/r12_a_energy_dfn.json:en  |
| Gravimetric energy density             | 493.2 Wh/kg             | cell/r12_a_energy_dfn.json:en  |
| Voltage plateau (discharge midpoint)   | 4.1301 V                | cell/r12_a_energy_dfn.json:mi  |
| Charge cut-off / discharge cut-off     | 4.7 V / 2.5 V           | LNMO system file               |
| DC internal resistance (1C, 10% point) | 6.87 mΩ                 | cell/r12_a_energy_dfn.json:dc  |
| Peak power density (theoretical)       | 12.25 kW/kg             | cell/r12_a_energy_dfn.json:po  |

### Fast charge (4C, 45 °C ambient)

| Item                             | Value                                        | Source                        |
|----------------------------------|----------------------------------------------|-------------------------------|
| 4C charge support                | Pass, no lithium plating (min anode potentia | cell/r12_a_4c45_dfn.json:anod |
| Max temperature during 4C charge | 49.55 °C (322.70 K) at 45 °C ambient         | cell/r12_a_4c45_dfn.json:T_ma |
| Note                             | Protocol current uses system-file nominal 4. | —                             |

### Life

| Item                                      | Value                                        | Source                          |
|-------------------------------------------|----------------------------------------------|---------------------------------|
| Anode SEI thickness after 100 × 1C cycles | 415.6 nm                                     | cell/r12_a_aging.json:sei_thick |
| SEI control                               | Dense FEC/VC-class film (EC diffusivity 5.0e | params r12_a_h140.json          |

### Mechanical / thermal interface

| Item                                  | Value                                        |
|---------------------------------------|----------------------------------------------|
| Stack thickness (Σ layers)            | 235.6 μm                                     |
| Electrode geometric area              | 0.1027 m²                                    |
| Cell mass                             | 53.77 g                                      |
| Cooling surface area (exposed)        | 0.0106 m² (double-sided thin pouch)          |
| Heat transfer coefficient requirement | ≥ 140 W/m²·K (vapor-chamber / chassis-couple |

### Construction

LNMO spinel cathode (density 4400 kg/m³, 74.75 μm, porosity 0.335, particle 3.0 μm) · graphite anode (120 μm, porosity 0.25, particle 1.5 μm) · 25 μm separator · EC/EMC + LiPF6 electrolyte (Chen2020 baseline) · Al/Cu current collectors.  
Layer areal masses: cathode 0.2212, anode 0.1491, Al CC 0.0432, Cu CC 0.1075, separator 0.0025 kg/m² (cell/r12\_a\_energy\_dfn.json:layer\_kg\_m2).

### Caveats

- LNMO file's deprecated positive-electrode diffusivity override is ignored by pybamm (positive particle diffusivity used instead) — disclosed, not compensated.
- Aging protocol per-cycle capacities are negative artifacts of the LNMO parameterization; SEI thickness is the reliable degradation indicator.
- True-compute endorsement skipped (real\_compute=false); values are DFN-simulation grade.